Modeling ELENA dipoles

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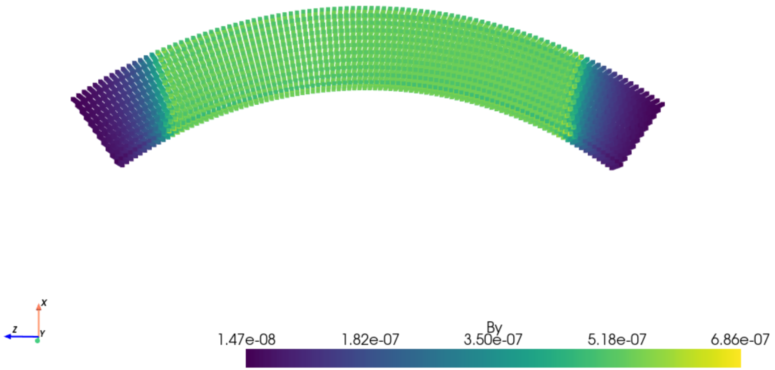
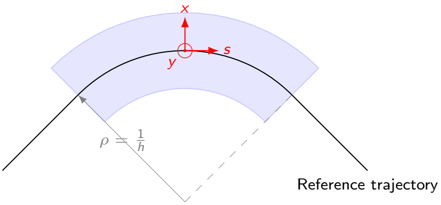
ELENA magnet

Fieldmaps from V. Ferrentino, M. Liebsch

```
data = np.loadtxt("ELENA_fieldmap.csv")
dipole = bpmeth.Fieldmap(data)
dipole.plot("By")
```



Reference frame for bent magnet



Scalar potential ($\mathbf{b} = -\nabla\phi$) has to satisfy Laplace equation in curved coordinates

$$\begin{aligned}\nabla \cdot (\nabla\phi) = & \frac{1}{1+hx}\partial_x\left((1+hx)\partial_x\phi\right) \\ & + \partial_y^2\phi + \frac{1}{1+hx}\partial_s\left(\frac{1}{1+hx}\partial_s\phi\right) \\ = & \ 0\end{aligned}$$

Solution of the Laplace equation

Scalar potential expanded in vertical plane

$$\phi = \sum_{i=0}^{\infty} \phi_i(x,s) \frac{y^i}{i!}$$

Solving Laplace equation gives recursion relation for ϕ_i

$$\phi_{i+2} = -\frac{1}{1+hx}\left(\partial_x\left((1+hx)\partial_x\phi_i\right) + \partial_s\left(\frac{1}{1+hx}\partial_s\phi_i\right)\right)$$

so two independent initial functions,

$$\phi_0(x,s) = -a_0(s) - \sum_{n=1}^{\infty} a_n(s)\frac{x^n}{n!}$$

$$\phi_1(x,s) = -\sum_{n=1}^{\infty} b_n(s)\frac{x^{n-1}}{(n-1)!}$$

with

$$b_n = \partial_x^{n-1} b_y\big|_{x=y=0} \qquad a_n = \partial_x^{n-1} b_x\big|_{x=y=0} \qquad a_0' := b_s = b_s\big|_{x=y=0}$$

$\Longrightarrow$ In the limit to s-independent fields and without curvature this reduces to the usual two dimensional multipole expansion

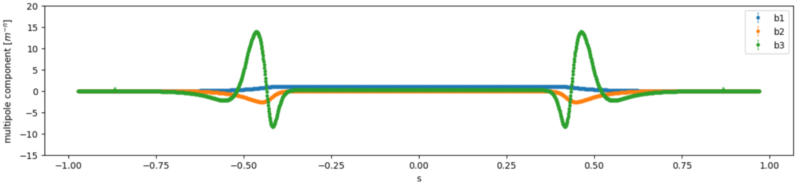
Field expansion

	1	x	y	x^2	xy	y^2
B_x	a_1	a_2	b_2	$\frac{a_3}{2}$	b_3	$\frac{-a_3-h(a_2-a_1h-2b_s')+b_sh'-a_1''}{2}$
B_y	b_1	b_2	$-b_s'-a_1h-a_2$	$\frac{b_3}{2}$	$-a_3-h(a_2-a_1h-2b_s')+b_sh'-a_1''$	$-\frac{b_3+hb_2+b_1''}{2}$
B_s	b_s	$-b_sh+a_1'$	b_1'	$b_sh^2-a_1'h+\frac{a_2'}{2}$	$-hb_1'+b_2'$	$-\frac{a_1h'+ha_1'+b_s''+a_2'}{2}$

ELENA components

- Fieldmap in Frenet-Serret coordinates
- Determine multipole components for each lonitudinal position

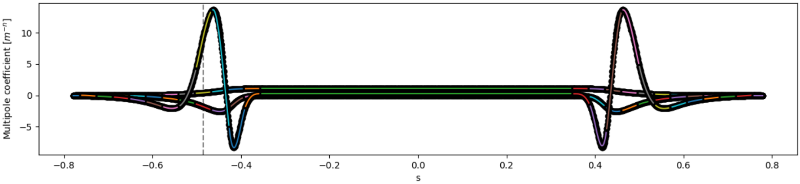
```
dipole_FS = dipole.calc_FS_coords(XFS=xvals, YFS=yvals, SFS=svals, rho=rho, phi=phi)
dipole_FS.z_multipoles(2)
```



Tracking through ELENA magnet

- Fit some function to longitudinal behaviour of components
  $\Longrightarrow$ Our choice: piecewise polynomials

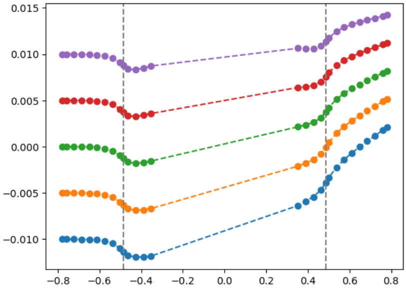
```
bpmeth.fit_segment(start_index, end_index, s_position, component, derivative_component)
```



Compensate closed orbit effect

- Build full magnet from segments
  - $\Longrightarrow$ We track with Runge-Kutta method
  - $\Longrightarrow$ Work of Sietse with Boris-like integrator
- Compensate closed orbit effect by rescaling magnet

```
bpmeth.PolySegment(length=length, h=h, comp=components)
```



Add element to Xsuite line

- Insert element in line as usual
- Include element in twiss by adding include_collective=True

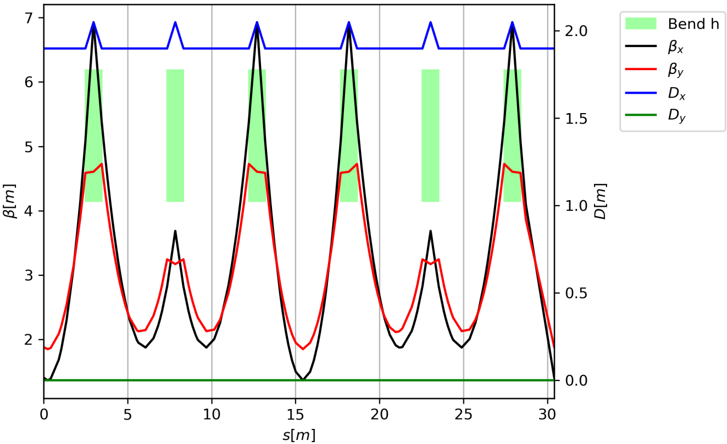
```
line.insert("fieldmap", polysegment, at=s_position)
line.twiss(include_collective=True)
```

Measurements in ELENA

ELENA measurements

Optics of ELENA are dominated by dipole edges

- Possible to turn off all quadrupoles in the machine
- Machine consists of only 6 main dipoles and dipole correctors



Impact on tune

Measurements 11/11/2025 - R. De Maria, D. Gamba, L. Ponce

Different models and parameters possible for description of dipole fringe field

Fitted θ_E to OPERA simulation

Fitted θ_E and F_{int} to solution of Hamilton equations with bpmeth

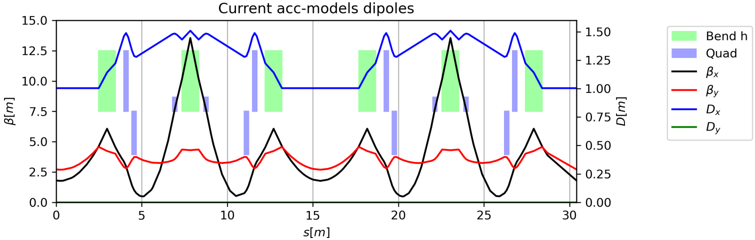
Solving Hamilton equations in fieldmap with bpmeth

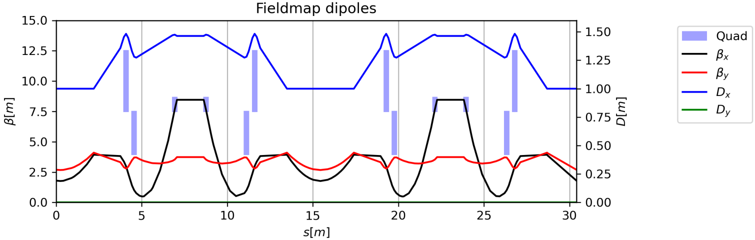
	Q_x	Q_y
Design parameters	1.7896	1.7708
Acc-models paramters	1.8362	1.7279
Fitted parameters	1.8406	1.7347
Fieldmap tracking	1.8364	1.7386
Measurement	x.8453	x.7386

- Xsuite fringe has two degrees of freedom, edge angle θ_E and fringe field integral F_{int}
- Parameters can be used to fit linear transfer matrix to solution of Hamilton equations in fieldmap

  $\Longrightarrow$ Tune fitting also possible
- Two parameters not sufficient to also fit non-linear behaviour: full solution (or better map) needed

Impact on optics

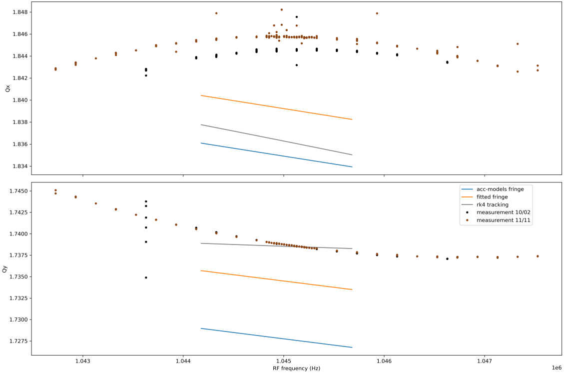




Chromaticity measurements

Measurements 11/11/2025 - R. De Maria, D. Gamba, L. Ponce

Currently unexplained second-order chromaticity

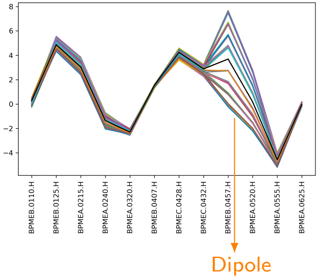


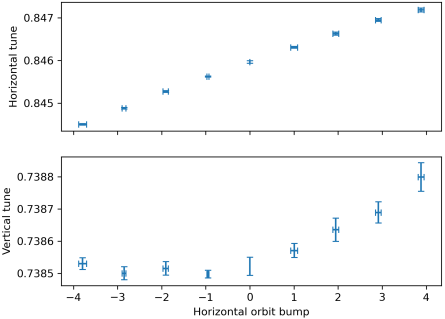
- Measurements with (11/11) and without (10/02) electron cooler have similar results
- Orbit in machine $\implies$ δ of measurements not exactly known

Orbit bump measurements

Measurements 10/02/2025 - R. De Maria, D. Gamba, J. Romain

Horizontal orbit bump inside a dipole



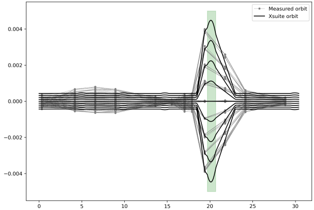


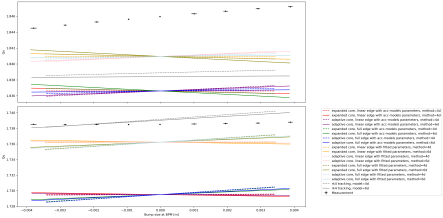
Orbit bump measurements

Comparison with simulation

Horizontal orbit bump inside a dipole: what plays a role?

- Change in δ + chromaticity
- Feeddown from dipole curvature terms
- Fringe field + edge angle effect

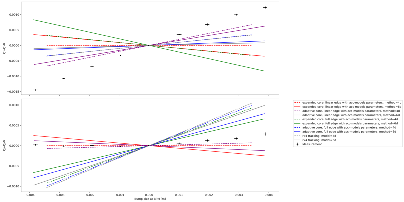


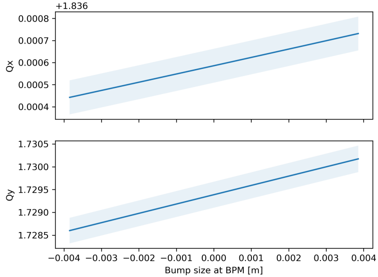


Orbit bump measurements

Comparison with simulation

- Left: Currently no good agreement between slope in model and measurement, but very sensitive
- Right: Orbit errors do not change the slope in simulation

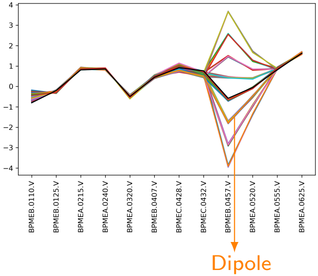


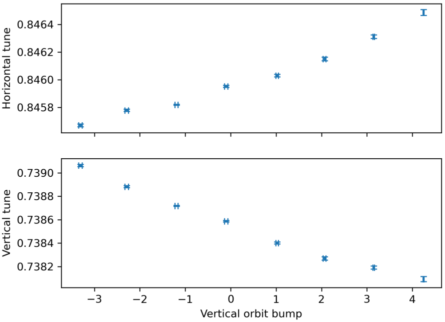


Orbit bump measurements

Vertical orbit bump inside a dipole

- To be compared with simulation





Conclusions

Short term:

- Analyse last MD data and reproduce results in simulation
- Apply resonance driving terms to HeLICS ring
- Collaborate on LEIR modelling

Long term:

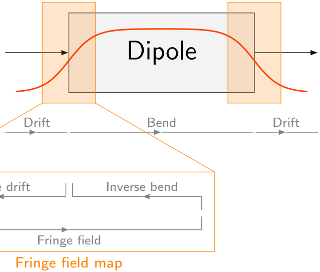
- Consolidate tracking method into Xsuite

Backup

Tracking fringe fields

Tracking fringe fields

- XSuite: hard edge map by E. Forest
- Resonance driving terms
- Runge-Kutta 4 method on Hamilton equations



$$x_f = x + \frac{1}{2}\frac{\partial\psi}{\partial p_x}{y_f}^2 \qquad p_{x,f} = p_x$$

$$y_f = \frac{2y}{1+\sqrt{1-2\frac{\partial\psi}{\partial p_y}y}} \qquad p_{y,f} = p_y - \psi y_f - \frac{b_0^2}{9Kg(1+\delta)}\,y_f^3$$

$$\ell_f = \ell - \frac{1}{2}\frac{\partial\psi}{\partial\delta}{y_f}^2 \qquad \delta_f = \delta$$

with

$$\psi = b_0 \tan\left[\arctan\left(\frac{x'}{1+y'^2}\right) - gb_0K\left(1+\frac{p_x^2}{p_s^2}\left(2+\frac{p_y^2}{p_s^2}\right)\right)p_s\right]$$

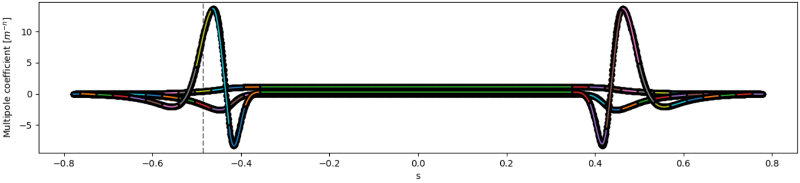
Tracking fringe fields

- XSuite: hard edge map by E. Forest
- Resonance driving terms
- Runge-Kutta 4 method on Hamilton equations

- $f_{1000}(s)$, $f_{0100}(s)$
- f_{10kl} and f_{01kl} with $k + l = 2n$
  - $n = 1$: f_{1020}, f_{1011}, f_{1002}, f_{0120}, f_{0111}, f_{0102}
  - $n = 2$: f_{1040}, f_{1031}, f_{1022}, f_{1013}, f_{1004}, f_{0140}, f_{0131}, f_{0122}, f_{0113}, f_{0104}
  - ...
- f_{00kl} with $k + l = 2(m + n)$
  - $m, n = 1$: f_{0040}, f_{0031}, f_{0013}, f_{0004}, h_{0022}
  - ...

Tracking fringe fields

- XSuite: hard edge map by E. Forest
- Resonance driving terms
- Runge-Kutta 4 method on Hamilton equations



Multipole expansion

Straight frame, no longitudinal dependence

The usual multipole expansion assumes an s-independent, and hence infinitely long magnetic field. In this case, one can apply two dimensional Maxwell equations, giving the scalar potential

$$\begin{aligned}\phi &= \mathcal{R}\left(\sum_{n=0}^{\infty} i\frac{(b_n + ia_n)}{n!}(x+iy)^n\right) \\ &= \mathcal{R}\left(\sum_{n=0}^{\infty} i\frac{c_n}{n!}r^n e^{in\theta}\right)\end{aligned}$$

with $c_n = b_n + ia_n$ or hence $b = -\nabla\phi$

$$b_y + ib_x = \sum_{n=0}^{\infty}\frac{(b_n + ia_n)}{(n-1)!}(x+iy)^{n-1}$$

with multipole components

$$b_n = \partial_x^{n-1} b_y\big|_{x=y=0} \qquad a_n = \partial_x^{n-1} b_x\big|_{x=y=0}$$

The coefficients c_n can be determined from a Fourier analysis of the data at a fixed radius (rotating coil measurement).

Generalized gradients

Straight frame, longitudinal dependence

If we want similar harmonic expansion of the data using a three dimensional field, we can propose a scalar potential with the same angular dependence

$$\phi = \mathcal{R} \sum_n \Phi_n(r, s) e^{in\theta}$$

Requiring to satisfy Maxwell equation now gives us the more general expansion

$$\phi = \mathcal{R} \left(\sum_{n=0}^{\infty} \frac{i}{n} e^{in\theta} \sum_{k=0}^{\infty} C_{n,k} \tilde{c}_n^{[2k]} r^{n+2k} \right)$$

with $C_{n,k} = \frac{(-1)^k}{2^{2k}} \frac{n!}{k!(k+n)!}$ a constant and $^{[\cdots]}$ representing derivatives with respect to s.
The coefficients $c_n^{[2k]}$ can be determined from a Fourier analysis of the data using multiple fixed radii.

Generalized gradients

Straight frame, longitudinal dependence: limit to s-independent field

Without s-dependence, the generalized gradient expansion

$$\phi = \mathcal{R}\left(\sum_{n=0}^{\infty}\frac{i}{n}e^{in\theta}\sum_{k=0}^{\infty}C_{n,k}\tilde{c}_n^{[2k]}r^{n+2k}\right)$$

reduces to

$$\phi = \mathcal{R}\left(\sum_{n=0}^{\infty}\frac{i}{n}e^{in\theta}\sum_{k=0}^{\infty}\tilde{c}_n r^n\right)$$

hence $c_n = (n-1)!\tilde{c}_n$

$\implies$ In the limit to s-independent fields this reduces to the usual two dimensional expansion up to a prefactor

Field derivatives on axis

Curved frame and longitudinal dependence

Scalar potential expanded in vertical plane

$$\phi = \sum_{i=0}^{\infty} \phi_i(x,s)\frac{y^i}{i!}$$

Maxwell equations give recursion relation for ϕ_i

$$\phi_{i+2} = -\frac{1}{1+hx}\left(\partial_x\left((1+hx)\partial_x\phi_i\right) + \partial_s\left(\frac{1}{1+hx}\partial_s\phi_i\right)\right)$$

so two independent initial functions,

$$\phi_0(x,s) = -a_0(s) - \sum_{n=1}^{\infty} a_n(s)\frac{x^n}{n!}$$

$$\phi_1(x,s) = -\sum_{n=1}^{\infty} b_n(s)\frac{x^{n-1}}{(n-1)!}$$

with

$$b_n = \partial_x^{n-1} b_y\big|_{x=y=0} \qquad a_n = \partial_x^{n-1} b_x\big|_{x=y=0} \qquad a_0' := b_s = b_s|_{x=y=0}$$

$\implies$ In the limit to s-independent fields and without curvature this reduces to the usual two dimensional multipole expansion

Field derivatives on axis

Curved frame and longitudinal dependence

This expansion is not harmonic in the angle!
First terms:

	1	x	y	x^2	xy	y^2
b_x	a_1	a_2	b_2	$\frac{a_3}{2}$	b_3	$\frac{-a_3-h(a_2-a_1h-2b_s')+b_sh'-a_1''}{2}$
b_y	b_1	b_2	$-b_s'-a_1h-a_2$	$\frac{b_3}{2}$	$-a_3-h(a_2-a_1h-2b_s')+b_sh'-a_1''$	$-\frac{b_3+hb_2+b_1''}{2}$
b_s	b_s	$-b_sh+a_1'$	b_1'	$b_sh^2-a_1'h+\frac{a_2'}{2}$	$-hb_1'+b_2'$	$-\frac{a_1h'+ha_1'+b_s''+a_2'}{2}$

Relation with generalized multipoles for straight frame $\Longrightarrow$ this restores the symmetry between the planes
(similar relation for skew terms)

$$b_n = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \frac{(-1)^k}{2^{2k}} \frac{(n-2k)!(n-1)!}{k!(n-k)!} \tilde{b}_{n-2k}^{[2k]}$$

Comparison first terms

Field derivatives VS generalized gradients

		1	x	y	x^2	xy	y^2
Field derivatives	b_x	a_1	a_2	b_2	$\frac{a_3}{2}$	b_3	$\frac{-a_3-a_1''}{2}$
	b_y	b_1	b_2	$-b_s'-a_2$	$\frac{b_3}{2}$	$-a_3-a_1''$	$-\frac{b_3+b_1''}{2}$
	b_s	b_s	a_1'	b_1'	$\frac{a_2'}{2}$	b_2'	$-\frac{b_s''+a_2'}{2}$
Generalized gradients	b_x	$\tilde{a}_1$	$\tilde{a}_2-\frac{\tilde{b}_s'}{2}$	$\tilde{b}_2$	$-\frac{3\tilde{a}_1''}{8}+\tilde{a}_3$	$-\frac{\tilde{b}_1''}{4}+2\tilde{b}_3$	$-\frac{\tilde{a}_1''}{8}-\tilde{a}_3$
	b_y	$\tilde{b}_1$	$\tilde{b}_2$	$-\tilde{a}_2-\frac{\tilde{b}_s'}{2}$	$-\frac{\tilde{b}_1''}{8}+\tilde{b}_3$	$-\frac{\tilde{a}_1''}{4}-2\tilde{a}_3$	$-\frac{3\tilde{b}_1''}{8}-\tilde{b}_3$
	b_s	$\tilde{b}_s$	$\tilde{a}_1'$	$\tilde{b}_1'$	$\frac{\tilde{a}_2'}{2}-\frac{\tilde{b}_s''}{4}$	$\tilde{b}_2'$	$-\frac{\tilde{a}_2'}{2}-\frac{\tilde{b}_s''}{4}$

Fringe field map for the dipole in PTC, XSuite, MAD-NG - Etienne Forest

- Derive the kick caused by an inverse drift - fringe - inverse bend on the particle
  - Lowest order $\Delta p_x = 0$
  - First order contribution Δp_y

$$\Delta p_{y,1} = -\frac{x'}{1+y'^2} b_0 y$$

  - Second order contribution Δp_y

$$\Delta p_{y,2} = \underbrace{\int_{-\infty}^{+\infty} b(s)(b_0 - b(s))ds}_{gb_0^2 K} \left(\frac{(1+\delta)^2 - p_y^2}{p_s^3} + \frac{p_x^2}{p_s^2} \frac{(1+\delta)^2 - p_x^2}{p_s^3} \right) y$$

- Construct a generating function that results in this kick

$$F = p_x x_f + p_y y_f - \delta \ell_f - \frac{1}{2} \psi(p_x, p_y, \delta) {y_f}^2$$

$$\psi = b_0 \tan \left[\arctan\left(\frac{x'}{1+y'^2} \right) - g b_0 K \left(1 + \frac{p_x^2}{p_s^2} \left(2 + \frac{p_y^2}{p_s^2} \right) \right) p_s \right]$$

- A map that is derived from a generating function is by definition symplectic

Forest, Leemann, Schmidt - 2006

Fringe field map for multipoles in PTC, XSuite, MAD-NG

- Hard edge multipole fringes: generating function

$$F(\mathbf{x}, \mathbf{p}^\mathbf{f}) = \mathbf{x} \cdot \mathbf{p}^\mathbf{f} \mp \mathcal{R} \frac{c_{n,body}\, r^n e^{in\phi}}{4(n+1)D} \left(r p_r^f + i \frac{n+2}{n} r p_\phi^f \right)$$

  - The Lee-Whiting quadrupole fringe

$$p_x = p_x^f \pm \frac{b_2}{12(1+\delta)} (3(x^2+y^2)\, p_x^f - 6xy\, p_y^f)$$
$$p_y = p_y^f \pm \frac{b_2}{12(1+\delta)} (6xy\, p_x^f - 3(x^2+y^2)\, p_y^f)$$

$\Longrightarrow$ Octupolar effect
$\Longrightarrow$ Reduces to sextupolar effect if there is an edge angle

- Quadrupole SAD soft fringe

- Missing: beta-beating originating from the finite extent of the fringe field

RDTs for fringe fields

$$f_{jklm} = \frac{\sum_w h_{w,jklm}\, e^{i[(j-k)\Delta\phi^{(s)}_{w,x}+(l-m)\Delta\phi^{(s)}_{w,y}]}}{\left(1 - e^{2\pi i[(j-k)Q_x+(l-m)Q_y]}\right)}$$

For a dipole fringe:

$$h_{1000}(s) = h_{0100}(s) = -((s) - \Theta(s))\,\frac{\sqrt{\beta_x(s)}}{2}$$

$$h_{10kl,k+l=2}(s) = \frac{\beta_y(s)(\alpha_x \pm i)}{8\sqrt{\beta_x}}\,\frac{(k+l)!}{k!l!}\,b_1'$$

$$h_{10kl,k+l=4}(s) = -\frac{\beta_y(s)^2(\alpha_x(s) \pm i)}{32\sqrt{\beta_x(s)}}\,\frac{(k+l)!}{k!l!}\,b_1'''(s)$$

$$h_{00kl,k+l=4}(s) = -\frac{3}{16}\,b_1'(s)^2\beta_y(s)^2\,\frac{1}{k!l!}$$